

ROBIN LANGE

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SUMMARY

Mechanical Engineering Honors student with research experience in distributed autonomy, multi-robot coordination, and experimental validation of autonomous systems. Interested in developing transferable planning and control methods for heterogeneous robots, particularly under uncertain or delayed information. Experience spans autonomy software, simulation, sensing, and mechanical hardware integration.

EDUCATION

B.S. Mechanical Engineering, Vehicle Design and Testing Emphasis

Georgia Southern University • Statesboro, Georgia • August 2023 – May 2027 • GPA: 4.00/4.00

- Honors College Scholar, President's List, Allen E. Paulson Scholar, Tau Beta Pi
- Honors thesis (in progress): When to Trust Learned Lead-Vehicle Predictions: Reliability-Gated Preview for Safety-Supervised Adaptive Cruise Control.
- Relevant coursework: Mechatronics, Mechanism Design, Machine Design, Dynamics, Mechanics of Materials, Measurement & Instrumentation Lab, Vehicle Design and Testing, Numerical Methods, Differential Equations, Calculus I–III, Program Design, Circuit Analysis, Logic Circuit Design, FEA, and Engineering Research.

EXPERIENCE

Undergraduate Student Researcher

Intelligent Vehicles Navigation Research Laboratory, Georgia Southern University

October 2023 – Present, Statesboro, GA

- Integrated ROS 2 visual SLAM, navigation, and behavior-tree mission logic on Jetson/RealSense hardware for indoor UAV research; measured 14.5 cm localization RMSE during a 110-second cart test over valid reference intervals.
- Developed a LiDAR-based hand-motion interface for UAV guidance, using point-cloud clustering and trajectory mapping to translate operator movements into robot commands.
- Investigating reliability-gated lead-vehicle prediction for safety-supervised model predictive adaptive cruise control; established a MATLAB/AVL PD commissioning baseline and designed matched comparisons of physical and learned preview.

Collaborative Robotics Autonomy Intern

National Center for Collaborative Autonomy, Institute for Human & Machine Cognition

May 2026 – August 2026, Pensacola, FL

- Architected and developed a modular, distributed Rust/Python autonomy platform for heterogeneous robot teams, integrating multi-agent decision-making, capability-based task allocation, and hardware-agnostic interfaces.
- Built physics-based simulation and hybrid digital twins with motion-capture integration, distributed state sharing, command validation, geofencing, and runtime safeguards for simulation and hardware evaluation.
- Validated multi-robot collection-and-delivery missions in 10+ hardware trials with two Boston Dynamics Spot robots; tested Clearpath Husky integration and implemented cross-platform CI/CD.

Vehicle Testing Research Intern

CARISSMA Institute of Safety in Future Mobility

July 2025 – August 2025, Ingolstadt, Germany

- Designed and constructed a truck-rear soft crash target for multimodal pre-crash testing, supporting research on early collision detection and occupant protection.
- Analyzed LiDAR and 4D-radar measurements to assess target detectability across covering and reflector configurations; supported experimental setup, data acquisition, and vehicle testing.
- Preparing a first-author manuscript on soft-target design and validation, combining sensor-technology literature review with analysis of field-test data.

Autonomous Systems and AI Intern

Fraunhofer Institute for Transportation and Infrastructure Systems

June 2024 – August 2024, Ingolstadt, Germany

- Implemented hybrid mission-autonomy software in Python/ROS 2 for UAVs, combining finite-state-machine transition guards with reactive behavior trees using `py_trees`.
- Compared legacy state-machine, behavior-tree, and hybrid implementations; benchmarked execution time and analyzed lines of code to characterize implementation tradeoffs.
- Verified mission sequencing and emergency interruption in Fraunhofer's in-house flight simulator; documented the architecture, evaluation methods, and test results.

PUBLICATIONS

Design and Validation of a Soft Crash Target for Multimodal Sensor-Based Truck Underride Testing

Manuscript in preparation.

• Lange R., Sezgin F., Inderst M., Sutton T., Lugner R., Brandmeier T., Soloiu V.

Data-Driven Calibration of Vehicle Simulations for Accurate Control Response in Development and Testing

Technical conference paper • SAE • 2026

• Soloiu V., Sutton T., Mehrzed S., Lange R., Zimmerman C., Peralta Lopez G., <https://doi.org/10.4271/2026-01-0058>

Warehouse Platooning Framework Using a Sensor-Rich Leader and Minimal Perception Follower

Preprint • SSRN • 2026

• Soloiu V., Sutton T., Lange R., Zimmerman C., <https://doi.org/10.2139/ssrn.7067358>

Development of a Rule-Based Lidar Lane-Keeping Assistance System for Autonomous Vehicles Applications

Technical conference paper • SAE • 2025

• Soloiu V., Mehrzed S., Kroeger L., Pierce K., Sutton T., Lange R., <https://doi.org/10.4271/2025-01-8025>

AWARDS & HONORS

Gulfstream Innovation, Engineering and Flight Excellence Awards — Third Place, Undergraduate

Georgia Southern University, Allen E. Paulson College of Engineering and Computing Research Symposium • 2026

Honorable Mention Engineering Division Student Research Video Competition

Council on Undergraduate Research (CUR), Engineering Division • 2026

Georgia Power Innovation Undergraduate Awards Second Place

Georgia Southern University, Allen E. Paulson College of Engineering and Computing Research Symposium • 2025

INVOLVEMENT

Honors Peer Mentor

Georgia Southern Honors College • August 2026 - Present

Physics Teaching Assistant and Tutor

Georgia Southern Physics Department • January 2025 - Present

Community Leadership and Service

Southern Leaders • August 2023 - Present

First-Year Experience Peer Leader

Georgia Southern First & Second Year Experience Department • August 2024 - December 2024

Programming for Engineers Laboratory Assistant

Georgia Southern Electrical Engineering Department • January 2024 - May 2024

SKILLS

Robotics & Autonomy: ROS 2, MAVROS, Ardupilot, Zenoh, Joint Activity Graphs, behavior trees/state machines, Nav2 path planning, coordinate transformations, runtime safeguards.

Hardware & Engineering: Boston Dynamics Spot, Clearpath Husky, iRobot Create3, NVIDIA Jetson AGX Orin/Nano, RealSense stereo camera, Ouster 3D LiDAR, Altos 4D Radar, OptiTrack/Motive, motor/encoder integration, LabVIEW, SolidWorks, ANSYS (FEA/CFD).

Perception & Learning: Visual SLAM, Isaac ROS, LiDAR/depth-camera processing, point-cloud clustering, computer vision, object detection, sensor fusion, YOLO, ONNX.

Control & Simulation: PID control, MPC (ongoing thesis), AVL Scenario Designer/Simulator, Gazebo, Bevy/Avian3D, MCAP/Foxglove.

Programming & Development: Rust, Python, MATLAB, C++/Arduino, Protocol Buffers, Git, Linux, CUDA, PyTorch, Docker, CI/CD (GitHub Actions/GitLab).